Technical Appendix

Data Cleaning + Analysis

Data Cleaning

Our analysis is stadium specific, so the first step of this project was to create a unique dataset for each stadium. In each stadium-specific data set, every row is a game and the columns reflect comprehensive research on stadium details, hourly weather data, and play-by-play game statistics.

Baseball statistics included in the data set are listed below (from Pybaseball Statcast and MLB Stats API):

- Game date
- Game time
- Runs scored
- Home runs
- Strike outs
- Walks
- Hit by pitches
- Home run to hits ratio

An additional day/night column was added using the game time variable.

The weather metrics include the following (From OpenMeteo):

- Temperature
- Precipitation
- Humidity
- Pressure
- Wind speed
- Wind direction

Stadium orientation, wind direction, and wind speed we used to create 3 new features for the projection of wind along the vectors from home plate to center field, left center field, and right center field. Larger magnitudes indicate higher wind speeds and positive signs mean wind was blowing towards the designated outfield marker.

Analysis

In lieu of recalculating the Statcast Park Factor itself, we sought to recreate a Park Factor formula based on the correlation between existing statistics and the park factor itself. Linear regression analysis found that over 0.9 of the variance was explained by runs and strikeouts, so these two statistics became the focus of our analysis (although others are also included in the heatmaps).

We built four different heat maps for each team that display the changes in baseball statistics as wind speed, air pressure, humidity, and temperature vary. Each heat map is divided into quintiles, each representing a fifth of the games played, with the leftmost being the lowest fifth.

On the heatmap, the value itself is the difference between the mean value in that bracket and the overall mean (so 0.5 would mean 0.5 more runs were scored than average at that ballpark). The heatmap is also color-coded to help observe potential overall linear trends. Below each value, a bootstrapped (B=1000) standard error is calculated at a 68% confidence level, intended to represent 1 standard deviation.

Model Design

Feature Selection

Wind was measured as three directional components: toward left-center, center, and right-center field. The left-center and right-center components each correlated with the center-field component at $r > 0.94$, so only the center-field component was retained. A wind-speed interaction term showed no significance across specifications and was excluded.

Reliable data on retractable roof status (open or closed at game time) is not available. As a result, weather features are applied uniformly across all parks, with no dampening for domed or partially covered stadiums.

Train/Test Split

Data spans the 2021 to 2025 seasons. The split is temporal by month to assess forward-looking validity and to isolate park effects from team strength. Months were assigned to training or test sets at a 70/30 ratio using random assignment with a fixed seed for reproducibility.

Model Specification

The model is ridge regression with park-specific weather sensitivities. It has three structural components:

1. Fixed effects for each weather variable capture the league-average effect across all parks.
2. Park indicator variables absorb baseline stadium differences (dimensions, foul territory, sight lines) that shift offensive output independent of weather conditions.
3. Park-by-weather interaction terms allow each park to carry its own weather coefficients, so the temperature effect at Coors Field can diverge from the temperature effect at Oracle Park.

Continuous features were standardized using median and interquartile range (RobustScaler) to reduce the influence of outliers and facilitate coefficient comparison. The ridge penalty (alpha) was selected via 5-fold cross-validation (RidgeCV) over a log-spaced grid. This approach addresses collinearity introduced by the interaction terms and reduces overfitting risk in parks with limited game samples.

Home-team and season fixed effects are excluded. Team quality is controlled by modeling away-team-only outcomes. Season-to-season variation remains in the residual rather than being absorbed by fixed intercepts.

Dashboard Integration

The trained model exports its coefficients to dashboard_params_v6.json. The JavaScript client (dashboard_model_v6.js) loads these parameters and computes predictions locally in the browser. The Python explainer (explainer_v6.py) provides component-level breakdowns used to generate the waterfall chart.

Dashboard Design

The dashboard is a browser-based interactive tool that allows users to explore weather-driven performance effects across all 30 MLB stadiums. Model coefficients are exported from the Python pipeline to dashboard_params_v6.json and loaded by the JavaScript client (dashboard_model_v6.js), which computes predictions locally in the browser without requiring a backend server.

The left sidebar is shared across both views and serves as the primary control panel. Users set weather conditions using four sliders—Temperature (°F), Wind Speed (mph), Wind Direction, and Humidity (%)---and toggle between Day and Night game types. A searchable stadium list displays all 30 MLB teams with their city and any relevant tags (e.g., ROOF for retractable-roof stadiums). Once conditions are configured, the user clicks Run Projection to generate results. A stadium photo is displayed in the upper-right corner for context.

Projections

On the Projections page, the central output is a large predicted park factor (e.g., 96.1), shown alongside its deviation from the stadium's historical average and the underlying model formula. A Predicted vs. Historical Park Factor bar chart displays annual park factors from 2021–2025, color-coded green or red relative to a dotted baseline, allowing users to see how the weather-adjusted prediction compares to observed historical values. Below this, two metric cards display Predicted Away Runs and Predicted Away Strikeouts, each with a weather adjustment value showing how much current conditions shift the prediction from the park baseline. At the bottom of the page, four weather distribution histograms (Temperature, Wind Speed, Humidity, Wind Direction) show where the user's selected conditions fall relative to the stadium's historical game-day distribution.

Analysis

On the Analysis page, the top section features a narrative stadium description summarizing the park's physical characteristics, typical weather environment, and known effects on play.

The primary visualization is a set of four heatmaps per stadium, each displaying how a key baseball statistic changes as one weather variable, temperature, wind speed, air pressure, or humidity, varies across quintiles. Each quintile represents one fifth of games played at that stadium, ordered from lowest to highest weather values. Cell values show the difference from the stadium's overall mean (e.g., +0.5 indicates 0.5 more runs than average), and each cell includes a bootstrapped standard error (B = 1,000, 68% confidence interval) to convey uncertainty. Cells are color-coded to help users identify directional trends across the distribution.

Users can navigate between stadiums using a stadium selector, allowing direct comparison of weather sensitivity profiles across parks. The dashboard is designed to make model outputs interpretable to a non-technical audience by grounding predictions in familiar baseball statistics and providing visual uncertainty estimates alongside each result.